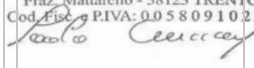


# UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

## TBM 13.08 M - CONSTRUCTION FOLLOW-UP

### IDENTIFICATION TABLE

|                  |                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Client           | PBDiM - Energopol                                                                                                                    |
| Project          | Udrożnienie Łódzkiego Węzła Kolejowego (TEN-T), Etap II, Odcinek Łódź Fabryczna - Łódź Kaliska/Łódź Żabieniec – Projekt POIIS 5.1-15 |
| Task             | TBM 13.08 m - CONSTRUCTION FOLLOW-UP                                                                                                 |
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| AUTHOR     |                                                                                      | REV                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opracował  | Miriam Polizzi                                                                       | 00                                                                                                                                                                                                               |
| Projektant | Paolo Cucino - Inżynierska mostowa i konstrukcyjno-budowlana, upr. nr ŁOD/TR/0247/19 | <b>SWS Engineering S.p.A.</b><br>Via della Stazione, 27<br>38123 Trento<br>Cod. Fisc. 00580910222<br>P.IVA: 00580910222<br> |

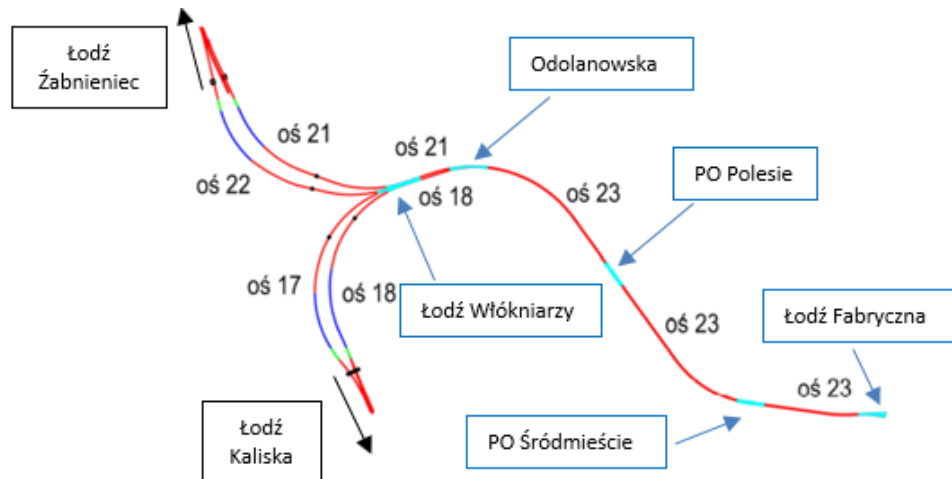
**INDEX**

|    |                                       |    |
|----|---------------------------------------|----|
| 1. | INTRODUCTION                          | 3  |
| 2. | FACE PRESSURES AND EXCAVATION VOLUMES | 4  |
| 3. | THRUST AND CUTTERHEAD TORQUE          | 6  |
| 4. | GROUT INJECTION                       | 7  |
| 5. | GUIDANCE                              | 8  |
| 6. | MONITORING                            | 9  |
| 7. | COMPENSATION GROUTING ACTIVITIES      | 14 |
| 8. | SOIL CONDITIONING                     | 15 |

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# 1. INTRODUCTION

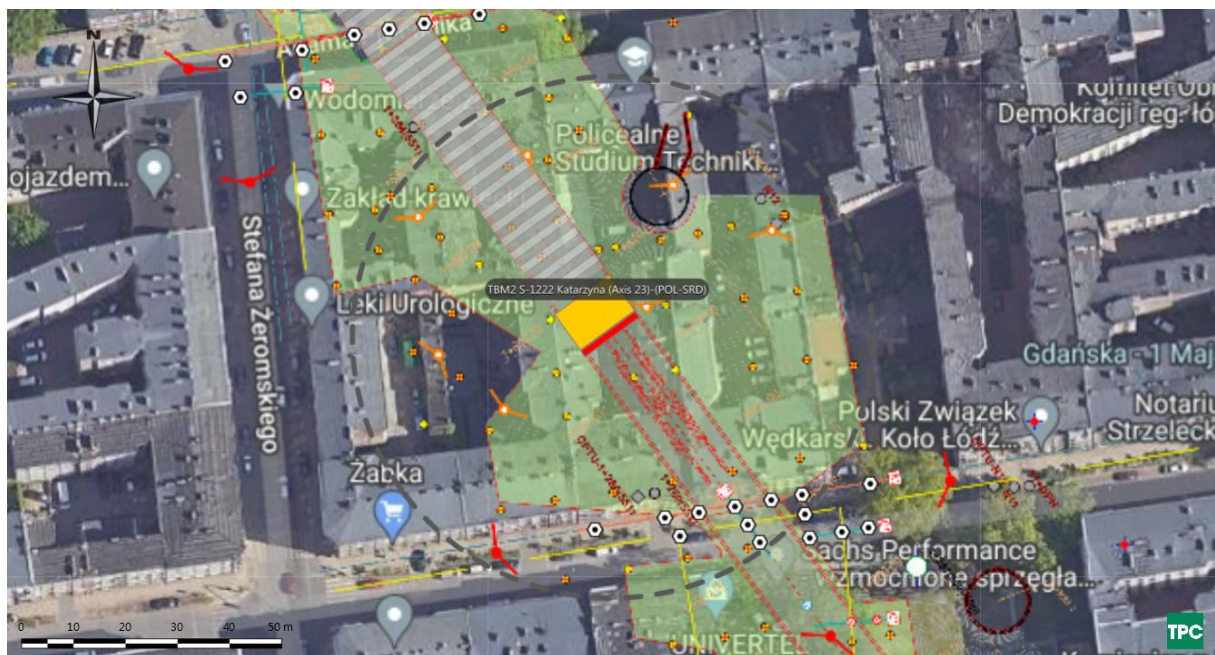
S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.



**Figure 1-1 Overview of main structural elements of the Project**

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 10/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+628.92 after the erection of ring n. 269; which corresponds to a total excavation length of 337.55 m from the starting chainage.

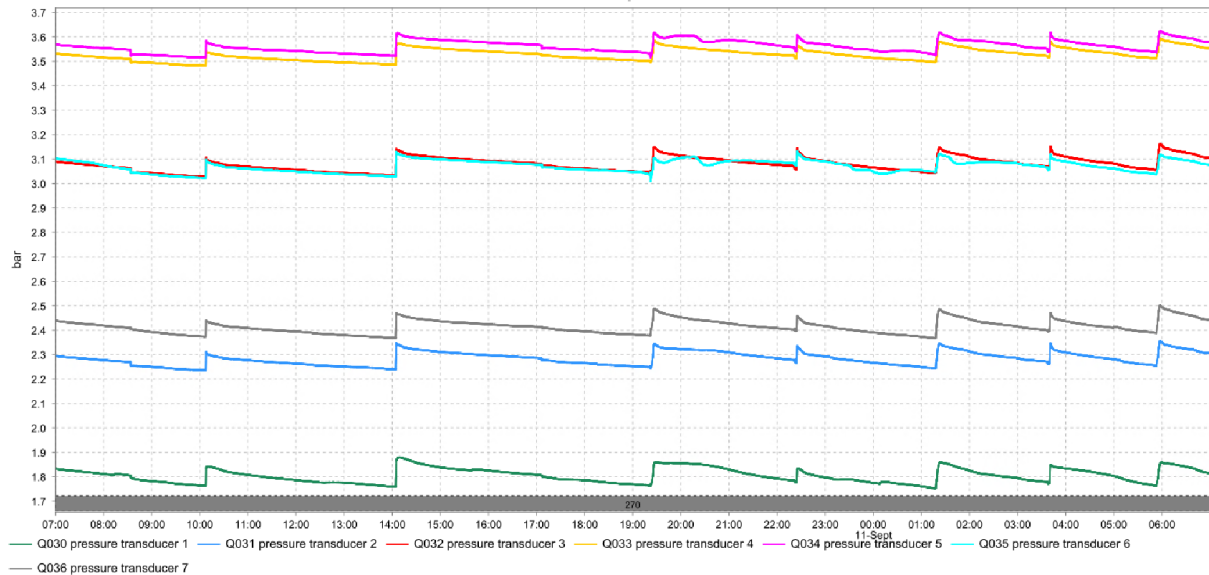


**Figure 1-2 Plan view of TBM Position on 11/09/2024**

## 2. FACE PRESSURES AND EXCAVATION VOLUMES

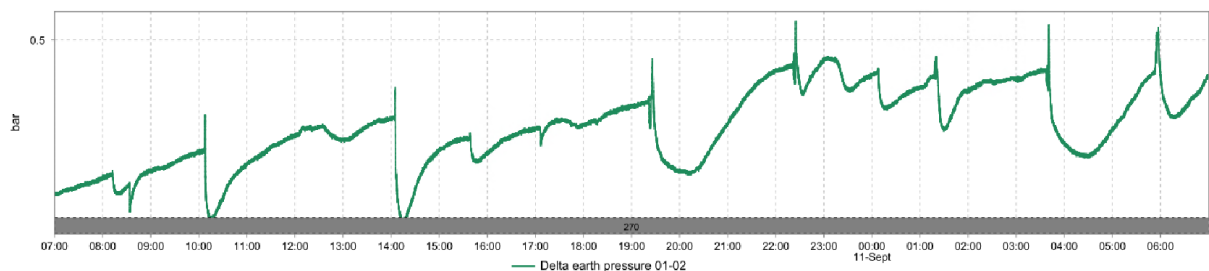
Average EPB pressure at the crown is at about 1.8 bar. The value is within the attention values.

Figure 2-1 shows EPB sensors.



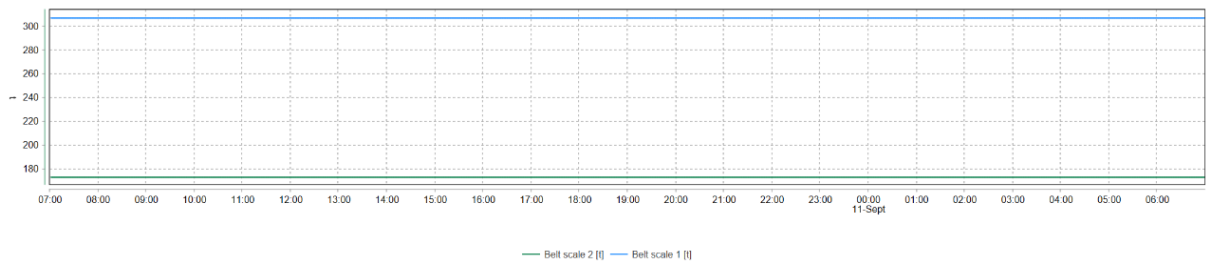
**Figure 2-1 EPB pressure sensor**

The delta between pressure transducer 1 and 2 has an average value of 0.48 bar indicating that the bulk chamber is filled. The observed increase in delta between pressure transducer 1 and 2 is most likely related to the gradual increase in the amount of bentonite injected into the excavation chamber.



**Figure 2-2 Delta between pressure sensors 1 and 2**

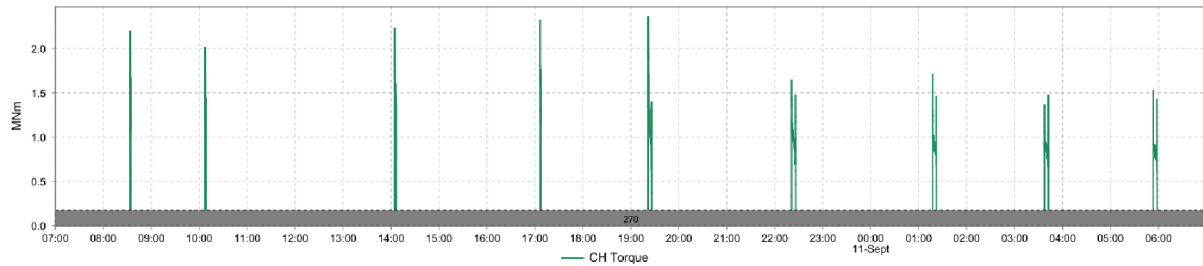
Belt scale n. 2 shows values lower than n. 1. It could be necessary the recalibration of belts before the restart of TBM excavation. The results of Belt scales have to be compared at the end of excavation of the ring. The belt scale value is a raw indication of the excavated material quantity.



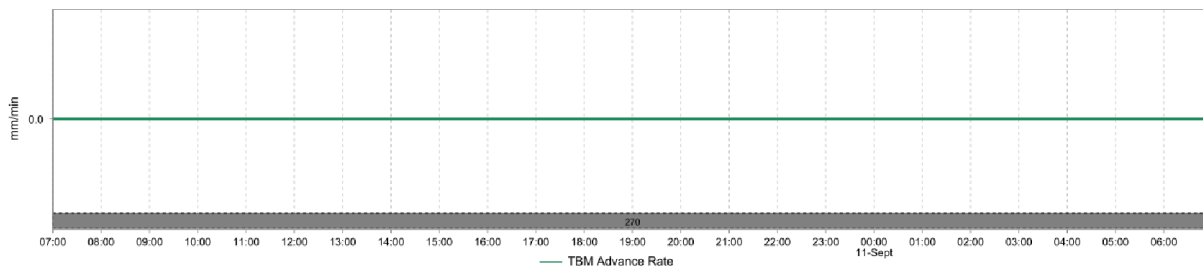
**Figure 2-3 Belt scales**

### 3. THRUST AND CUTTERHEAD TORQUE

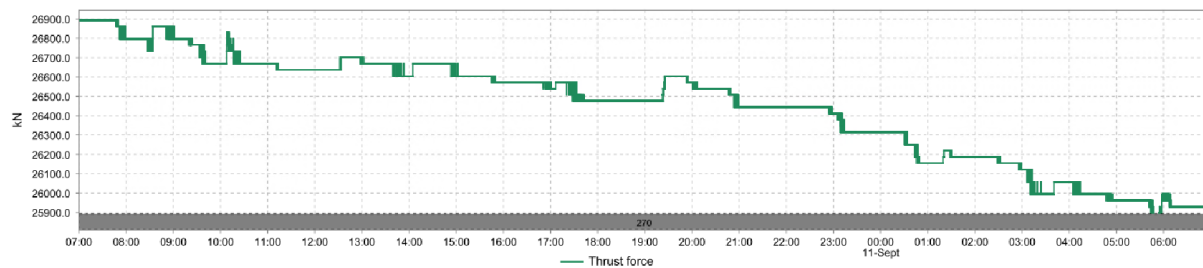
Graph for torque, advance rate and thrust force are below shown. The torque graph shows non-zero values due to the rotation of the cutterhead.



**Figure 3-1 CH Torque**



**Figure 3-2 Advance rate**



**Figure 3-3 Thrust force**

4. GROUT INJECTION

Grout injection volume has to be compared to minimum theoretical annular volume +10% of 13.54 m<sup>3</sup> at the end of ring excavation. Ratio B/A is at 5%. It was noted that peaks observed for average injection pressure lines are related to cleaning processes.

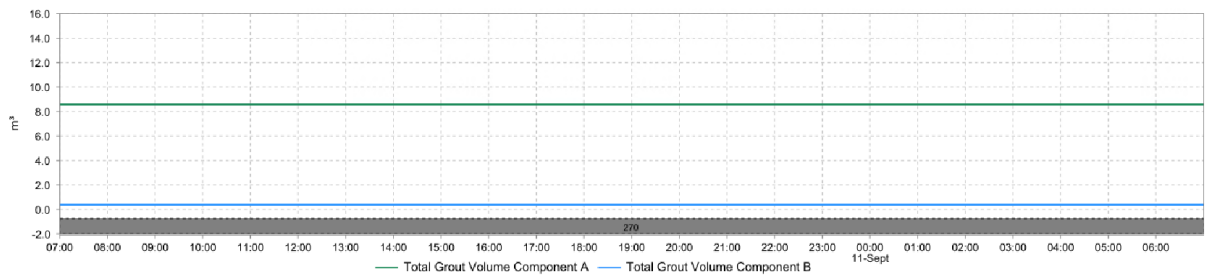


Figure 4-1 Grout volumes

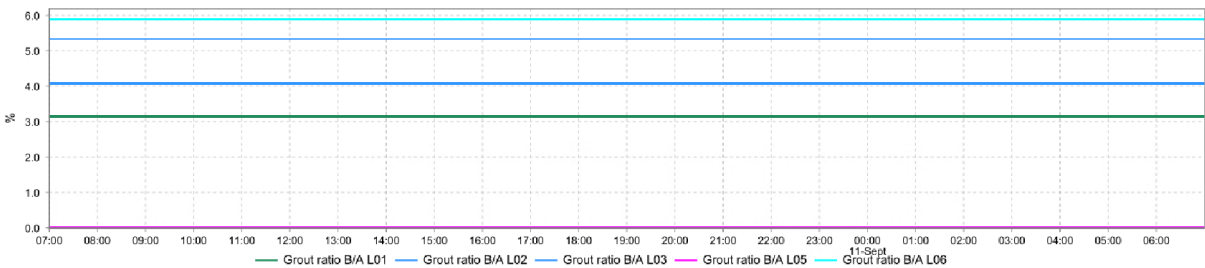


Figure 4-2 Volume ratio Components B/A

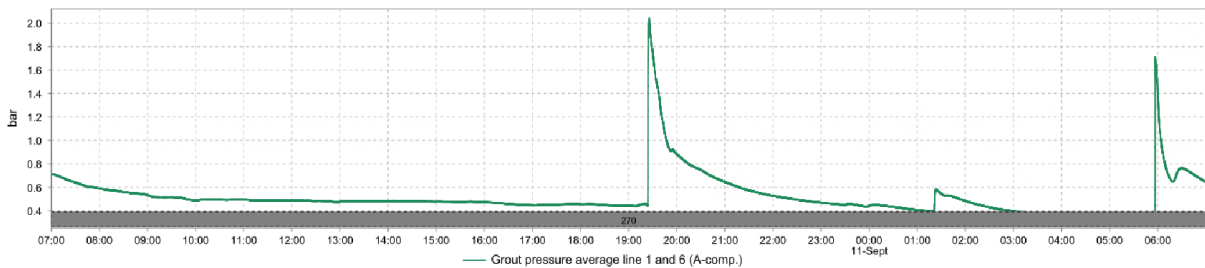


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -15.85 mm on the front edge of the shield and the horizontal deviation of about -17.86 mm (Figure 5-1). Maximum radial deviation is of 23.88 mm.

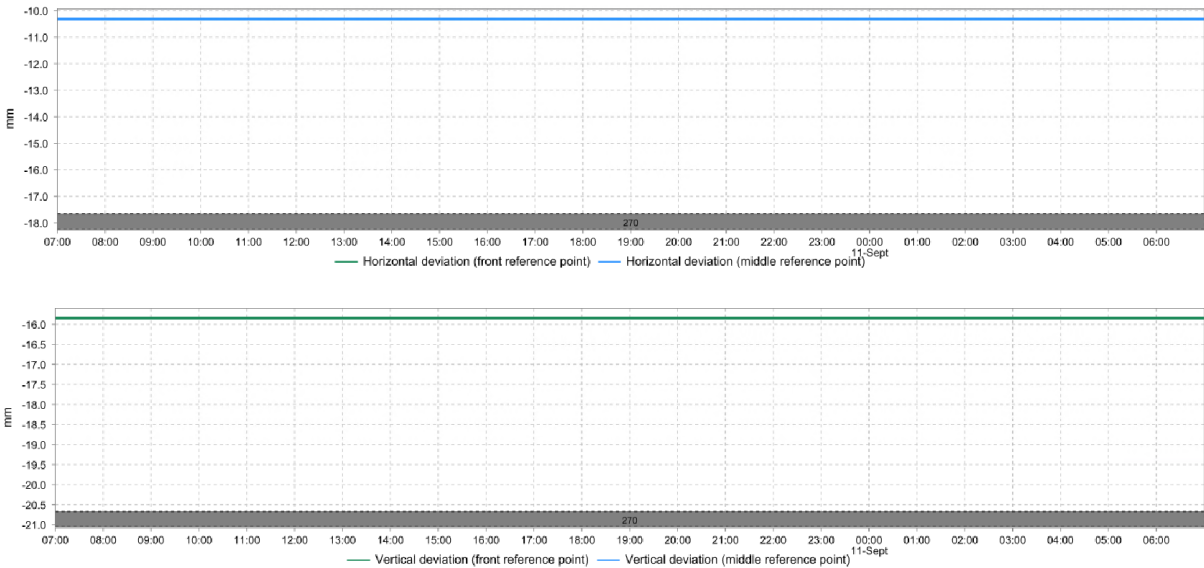


Figure 5-1 Machine deviation from the alignment



## 6. MONITORING

Based on data from Geoscope and Beyond, building subsidence has been observed but is showing signs of stabilization.



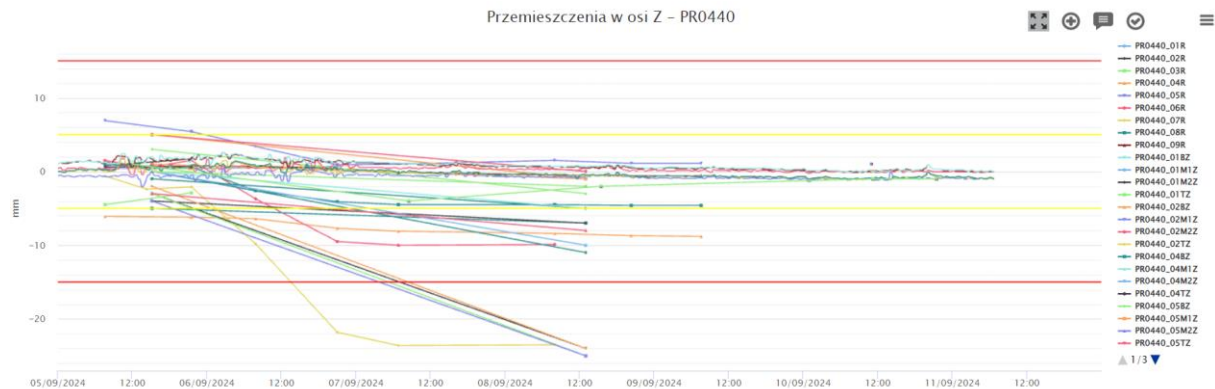
Figure 6-1 Monitoring points – Geoscope



Figure 6-2 Monitoring points – Beyond

On PR0440, Geoscope monitoring points show stabilizing trends starting from 07/09/2024 at 8:00 am, following the excavation stoppage. It is not possible to appreciate a complete stabilization trend because many points have not recorded new measurements since 08/09/2024 in the morning.

In Beyond, monitoring points show a small downward trend after an uplift.



**Figure 6-3 Settlement of building PR0440 - Geoscope**



**Figure 6-4 Monitoring point PR0440 - Beyond**

On PR0460, Geoscope monitoring points show downward trends. Points PR0460\_07M2Z, PR0460\_07TZ, PR0460\_A\_13TZ, PR0460\_A\_13BZ and PR0460\_A\_13M2Z have exceeded the attention threshold. Point PR0460\_09 showed a small uplift trend followed by a stabilizing trend.

In Beyond, monitoring points show downward trend for points PR0460\_07\_18L and PR0460\_13\_19L, that have exceeded the attention threshold. Points PR0460\_06\_18L and PR0460\_09\_18L showed a small uplift trend followed by a stabilizing trend.

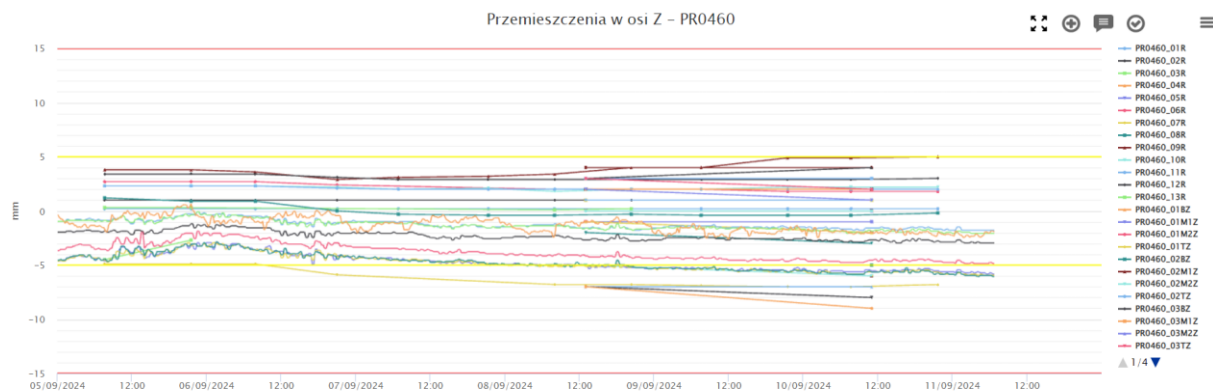


Figure 6-5 Settlement of building PR0460 - Geoscope



Figure 6-6 Monitoring point PR0460 - Beyond

On MA0190, monitoring points are stable.

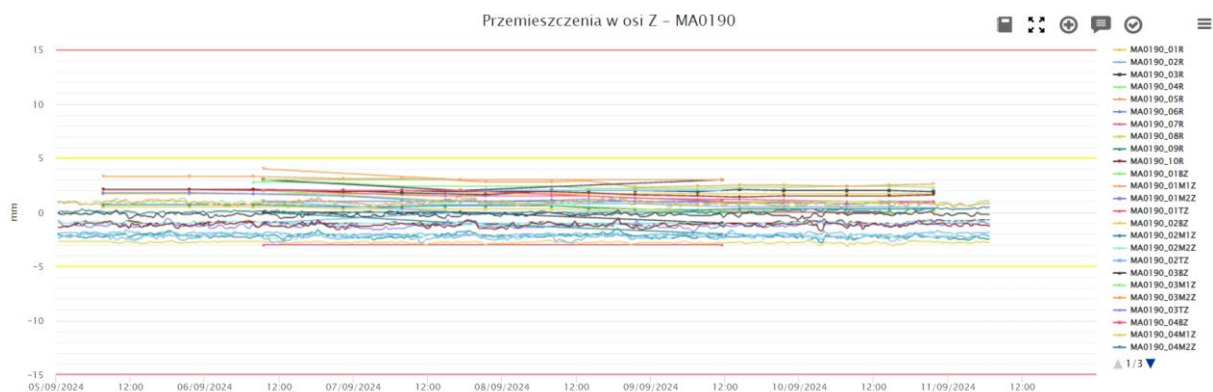


Figure 6-7 Settlement of building MA0190 - Geoscope

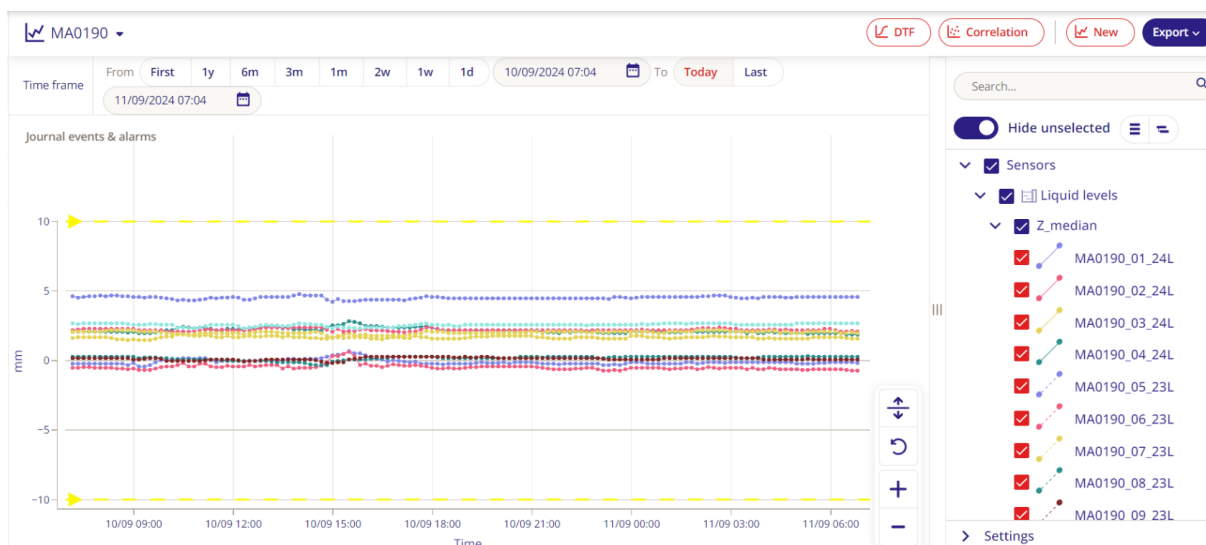


Figure 6-8 Monitoring point MA0190 - Beyond

On MA0210, Geoscope monitoring points show uplift trends in the last 24h.

In Beyond, monitoring points show stabilizing trends, except for MA0210\_05\_22L which showed a small uplift followed by a new downward trend.

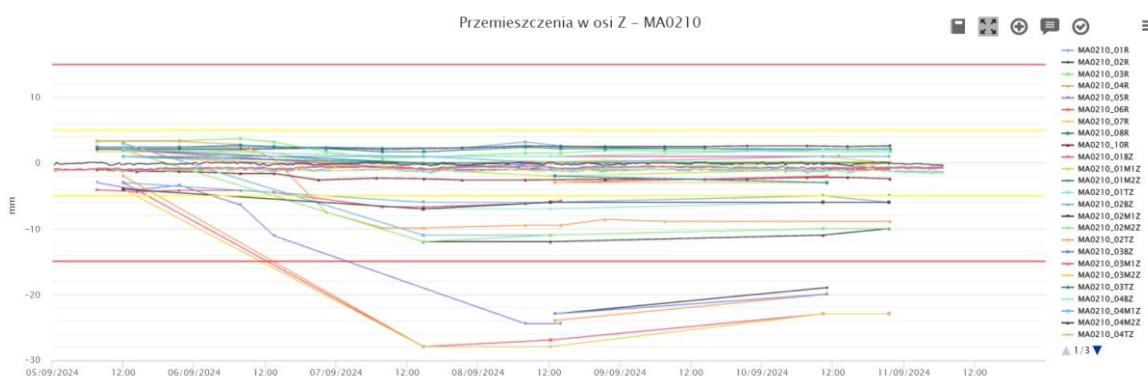


Figure 6-9 Settlement of building MA0210 - Geoscope

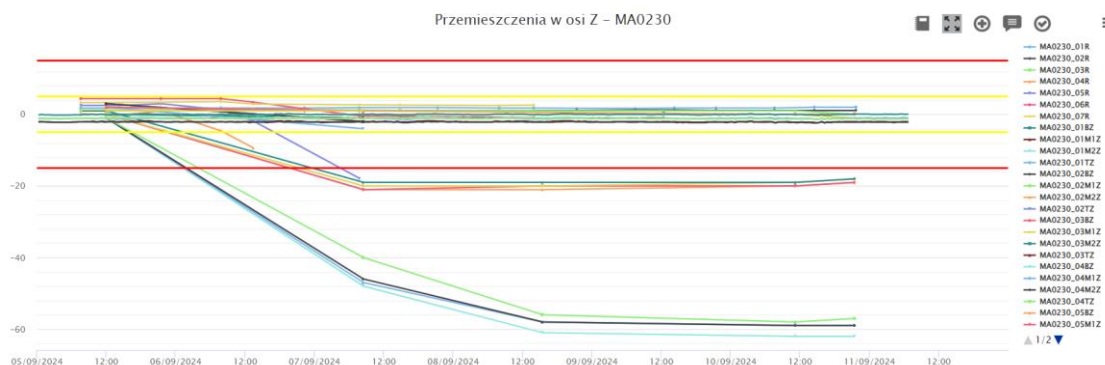


Figure 6-10 Monitoring point MA0210 - Beyond



On MA0230, Geoscope monitoring points show stabilizing trends following the excavation stoppage. Points MA0230\_04TZ, MA0230\_05BZ, MA0230\_05M1Z, MA0230\_05M2Z and MA0230\_05TZ show small uplift. Some points have not recorded new measurements since 06/09/2024 in the morning.

In Beyond, monitoring points show stabilizing trends.

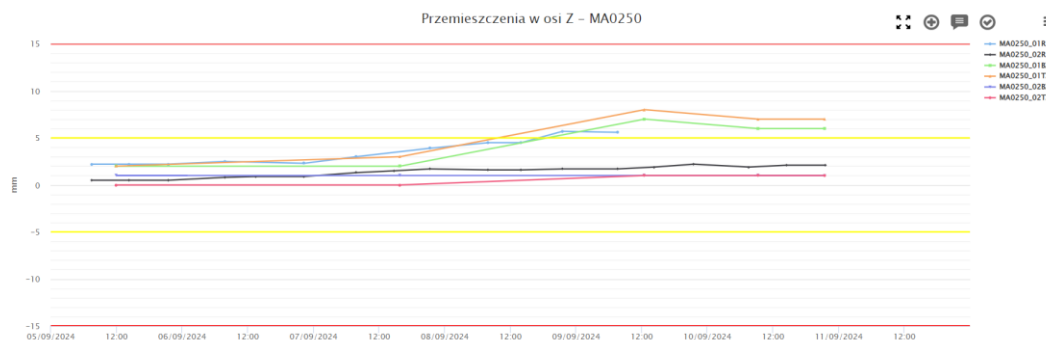


**Figure 6-11 Settlement of building MA0230 - Geoscope**



**Figure 6-12 Monitoring point MA0230 - Beyond**

On MA0250, monitoring points showed an uplift trend, followed first by a downward trend and then by a stabilizing trend. Points MA0250\_01BZ, MA0250\_02BZ and MA0250\_01TZ exceeded the attention threshold.



**Figure 6-13 Settlement of building MA0250 - Geoscope**

## 7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 10/09/2024 of Soletanche, compensation grouting has been performed on the 10/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 8610.55 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure.

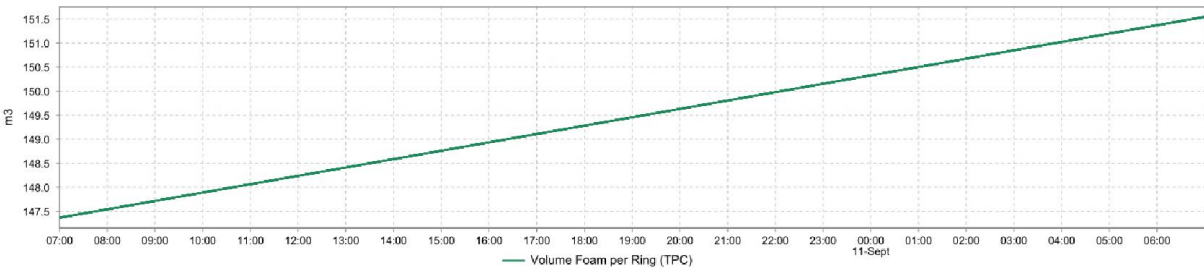


Figure 8-1 Injected foam volume

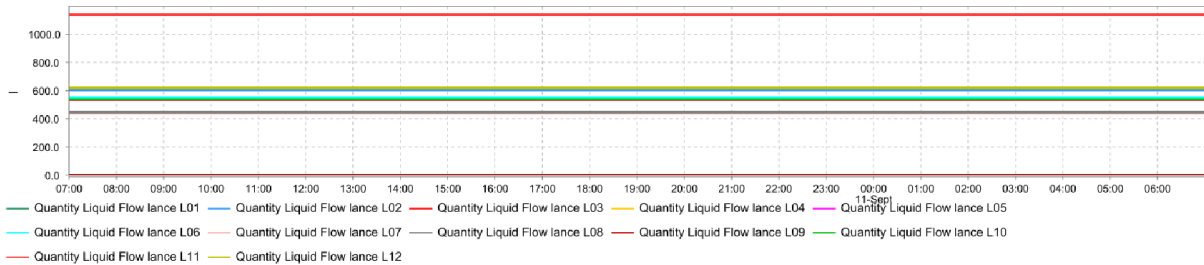


Figure 8-2 Conditioning liquid volume